Wenhan Yang

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Research Summary

My research focuses on multimodal foundation models. It spans data curation, post-training (SFT, RL, and distillation), defenses against poisoning and backdoor attacks, and efficient training and test-time inference. More recently, I train lightweight orchestrators that decompose long-horizon tasks and delegate to sub-agents, extending the effective execution horizon at a fraction of frontier-model cost.

Education

2022 – Present **Ph.D. in Computer Science, University of California, Los Angeles**
Advisor: Baharan Mirzasoleiman; Expected: June 2027

2019 – 2022 **B.S. Mathematics of Computation, University of California, Los Angeles**
*Minor in **Statistics**; Magna Cum Laude; Dean's Honors List; GPA: 3.941/4.0*

Research Publications

* Equal contribution

1. H. H. Jin*, **Yang, Wenhan***, M. Ghaffari, C. Morato, and B. Mirzasoleiman, "From weeks to hours: Fast and principled SFT curation for LLM," in *The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2026.
2. H. H. Jin*, **Yang, Wenhan***, M. Ghaffari, C. Morato, and B. Mirzasoleiman, "Reasoning quality emerges early: Data curation for reasoning models," in *The Forty-Third International Conference on Machine Learning (ICML)*, 2026.
3. **Yang, Wenhan**, N. Naharas, A. Payani, and B. Mirzasoleiman, "Distilling visual reasoning into text space," *Under review; arXiv:2609.34408*, 2026.
4. **Yang, Wenhan**, A. Rao, and A. Chandra, "HAWK: Rethinking multimodal drafting for speculative decoding," *Under review; arXiv:2610.00623*, 2026.
5. D. Nguyen*, **Yang, Wenhan***, R. Anand, Y. Yang, and B. Mirzasoleiman, "Memory-efficient training of LLMs with larger mini-batches," in *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025.
6. **Yang, Wenhan**, S. Stice, A. Payani, and B. Mirzasoleiman, "Bootstrapping LLM robustness for VLM safety via reducing the pretraining modality gap," *Under review; arXiv:2505.24208*, 2025.
7. **Yang, Wenhan**, J. Gao, and B. Mirzasoleiman, "Better safe than sorry: Pre-training CLIP against targeted data poisoning and backdoor attacks," in *The Forty-First International Conference on Machine Learning (ICML)*, 2024.
8. **Yang, Wenhan** and B. Mirzasoleiman, "Graph contrastive learning under heterophily via graph filters," in *The Fortieth Conference on Uncertainty in Artificial Intelligence (UAI)*, 2024.
9. S. Joshi, Y. Yang, Y. Xue, **Yang, Wenhan**, and B. Mirzasoleiman, "Towards mitigating spurious correlations in the wild: A benchmark & a more realistic dataset," *Under review; arXiv:2306.11957*, 2023.
10. **Yang, Wenhan**, J. Gao, and B. Mirzasoleiman, "Robust contrastive language-image pretraining against data poisoning and backdoor attacks," in *The Thirty-Seventh Annual Conference on Neural Information Processing Systems (NeurIPS)*, Effectively protects CLIP pre-training against up to 1% poison rate, 2023.

Employment History

Research Scientist Intern, **Samsung Research America (Visual Display Intelligence Lab)** Jun – Sep 2026

- **Research focus:** Developing multimodal drafting methods for lossless speculative decoding in VLMs.
- Proposed HAWK, a target-aware drafter that combines compressed visual states with features from selected target-model layers, improving access to visual information while reducing redundant computation.
- Introduced a dynamic logit-shifting loss that adjusts supervision when draft and target predictions disagree, improving draft–target alignment during multi-step decoding.
- Evaluated across 10 multimodal benchmarks, achieving 2.60× lossless speedup versus 2.19× for EAGLE-3 and increasing average acceptance length by 23%.

Student Researcher, **Google Research (Algorithms & Optimization)** Jan – Jun 2026

- **Research focus:** Post-training small language models for planning over long-horizon agentic trajectories.
- Investigated compact planners that decompose complex tasks and coordinate workers in multi-agent systems, targeting capabilities otherwise provided by much larger language models or expensive explicit search.
- Explored extensions to multi-turn diagnostic agents, where plans must incorporate new observations and adapt over long interaction histories.

Research Scientist Intern, **Cisco Research, Outshift (AI & ML)** Jun – Sep 2025

- **Research focus:** Transferring visual reasoning capabilities into textual chain-of-thought without generating intermediate images at inference time.
- Proposed V2T, a teacher–student framework that distills interleaved visual–textual reasoning into a VLM using textual supervision, logit distillation, and attention alignment.
- Developed a two-stage training pipeline combining supervised distillation with GRPO, incorporating task correctness and visual grounding into the reward.
- Improved average accuracy by 14.3% on held-out tasks and 2.7% on a broader evaluation suite, outperforming the teacher and competing baselines.

Research Scientist Intern, **Samsung Research America (NLU & On-Device AI Lab)** Jun – Sep 2024

- **Research focus:** Developing adaptive audio-token compression for efficient on-device speech LLMs.
- Extended the Matryoshka Query Transformer to SALMONN to represent audio inputs using nested query-token subsets, supporting multiple compression levels within a single model.
- Introduced a learned, input-adaptive token budget to allocate audio tokens according to the input, enabling variable inference costs without training separate models for each token budget.

Awards & Leadership

2025 **Summer Mentored Research Award,** University of California, Los Angeles

2023 – 2025 **President,** DanceSport Club at UCLA

Academic Service

Conference Reviewer NeurIPS, ICML (2023–26); ICLR (2023, 2026–27); AISTATS, UAI (2025–26)

Journal Reviewer Transactions on Machine Learning Research (TMLR)

Workshop Reviewer AdvML-Frontiers @ ICML 2023, AdvML-Frontiers @ NeurIPS 2024

Teaching Experience

Teaching Assistant

CS 260D: Large-Scale Machine Learning, UCLA Fall 2023, Fall 2025, Fall 2026

CS M148: Introduction to Data Science, UCLA Winter 2023, Winter 2025

CS 188: Introduction to Human-Computer Interaction, UCLA Spring 2025